

ANALOG & DIGITAL CIRCUITS

Lecture 5

Computer Science Department,
TUCN

NMOS Digital Circuits

Introduction

Static NMOS circuits

- Example of a Dynamic NMOS circuit

CMOS Logic Integrated Circuits

Introduction

CMOS Inverter

Parameters of CMOS circuits

Circuits for protection

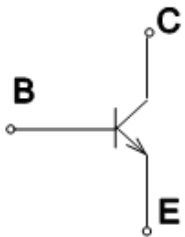
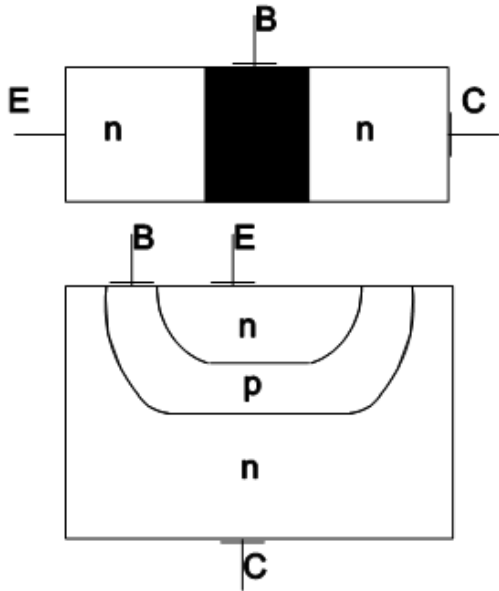
Output stage for CMOS circuits

Buffering circuits

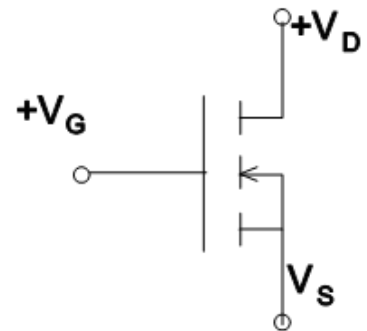
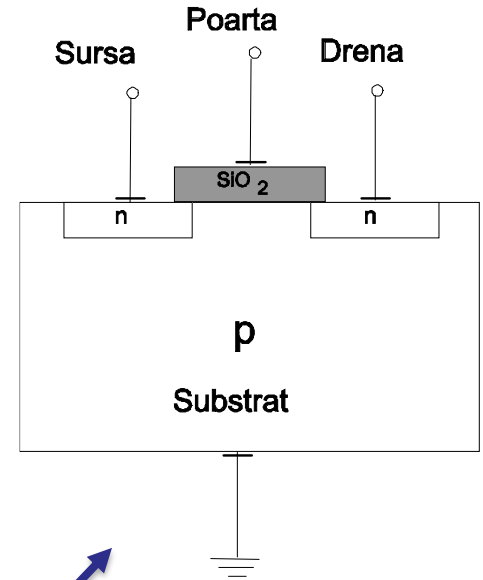
Introduction

- PMOS and NMOS families are based on MOS transistors with induced channel of **p**, respectively **n**
- NMOS circuits mostly used for switching (higher speed)
- Circuits are using exclusively NMOS transistors
- One positive power supply
- Logic levels are function of power supply voltage level

Bipolar transistor vs MOS transistor

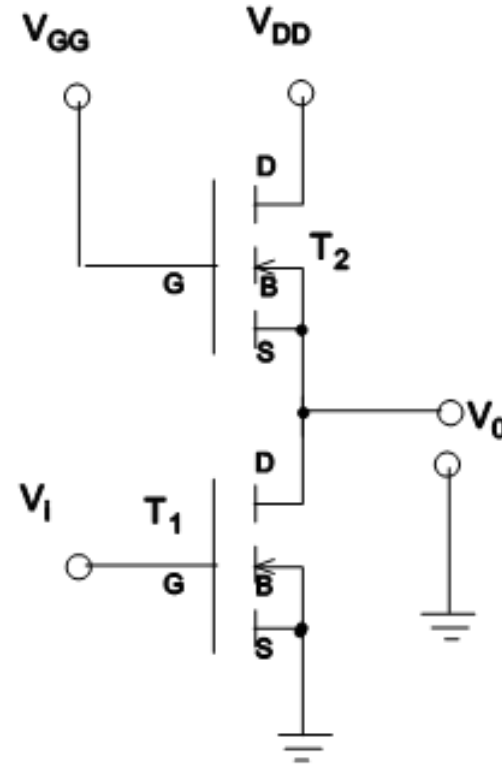
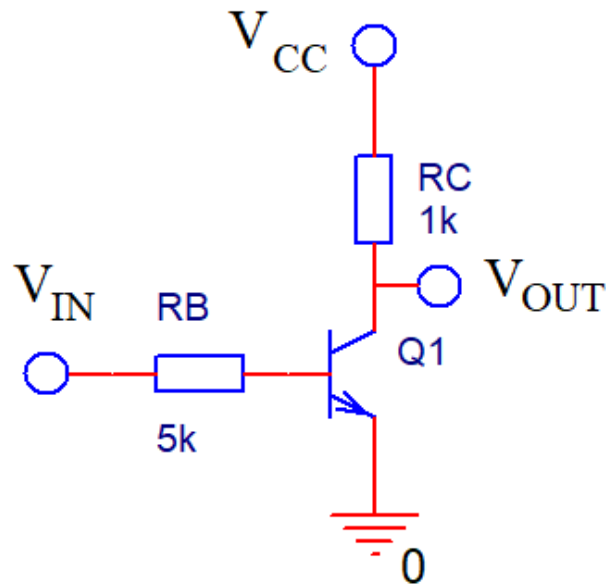


- At BJT we have V_{BE} and I_B controlling the functioning $\Rightarrow V_{CE}$;
- Need an R_B to control I_B



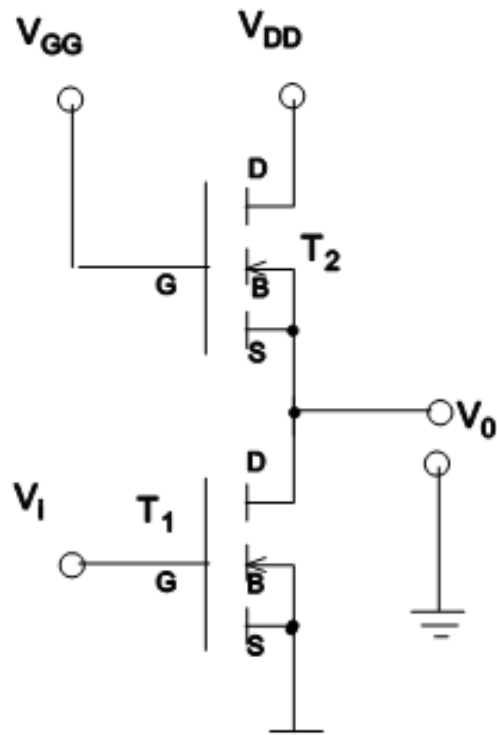
- At MOS we have V_{GS} controlling the functioning (I_g current is negligible ~ 0)
- Drive V_{DS} by field effect, not by a current $\Rightarrow$ no need for a resistor since the current is ~ 0

Static NMOS inverter



- Transistor T_1 acts as an inverter
- T_2 acts as active load, replacing the static resistor
- MOS technology use transistors as playing resistor role
- T_1 is based on **n** channel, in enhancement mode; T_2 works in depletion mode
 - It means: T_1 threshold voltage positive and T_2 being negative
- External load is normally also NMOS inputs, so there is a huge input resistance and the load has mainly a capacitance meaning

Transfer Characteristic



Area a, $V_i < V_{T1}$, T1 is off, $I_{DS1} = 0$, $V_o = V_{DD}$, T₂ works in linear regime

Area b, $V_i > V_{T1}$, T2 works in linear region. $V_{DS1} > V_{GS1} - V_{T1}$, T₁ saturated. For T₂ being in linear region, $0 \leq V_{DS2} \leq V_{GS2} - V_{T2}$, where $V_{GS2} = 0$, and $V_{DS2} = V_{DD} - V_o$, therefore V_o must be higher than $V_{DD} + V_{T2}$, making at input:

$$V_{T1} < V_i < V_{T1}(1 + \sqrt{2})$$

Area c

$$V_i = V_{T1}(1 + \sqrt{2})$$

T2 saturated

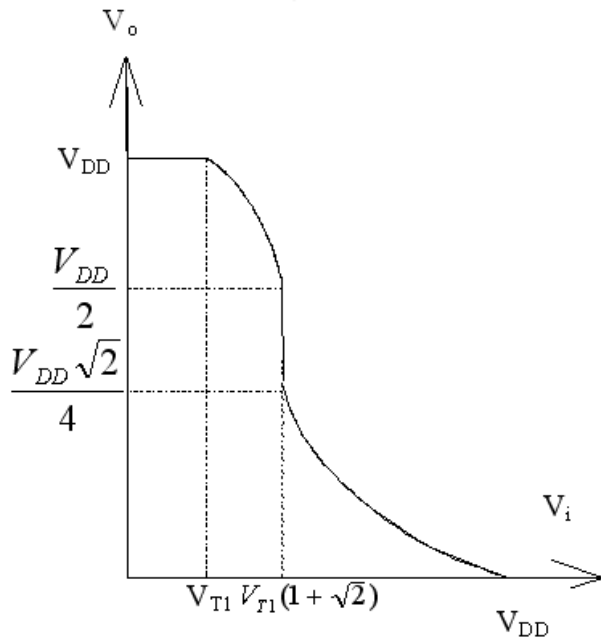
T1 saturated for:

$$\frac{V_{DD}\sqrt{2}}{4} \leq V_o \leq \frac{V_{DD}}{2}$$

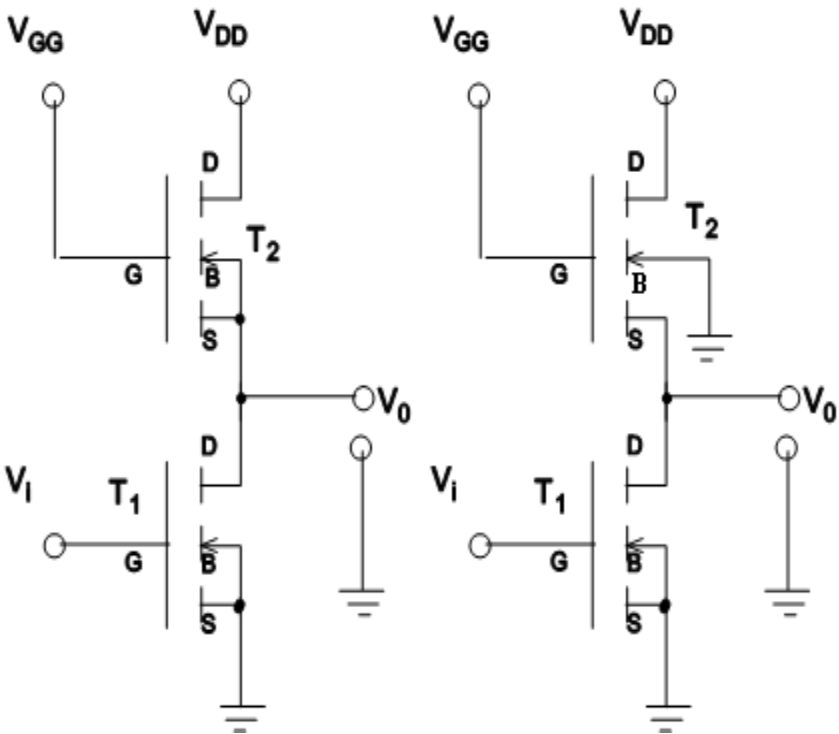
Here transfer characteristic is linear and abrupt

Area d, T₁ off saturation and enter linear regime; T₁ goes off saturation for:

$$V_i \approx \frac{V_{DD}^2}{16} \cdot \frac{1}{V_o} + \frac{V_{DD}}{8} + \frac{V_o}{4}$$



Load resistance



Built using a transistor: T_2

Gate connected to V_{GG}

T_1 on, for having V_o very close to zero, $R_{T2} \gg R_{T1}$:

$$\frac{W_1/L_1}{W_2/L_2} \gg 1$$

R_{T1} has values: 0,5 up to 10K Ω

If $R_{T1}=10\text{K}\Omega$ then $R_{T2}=250\text{K}\Omega=R_s$

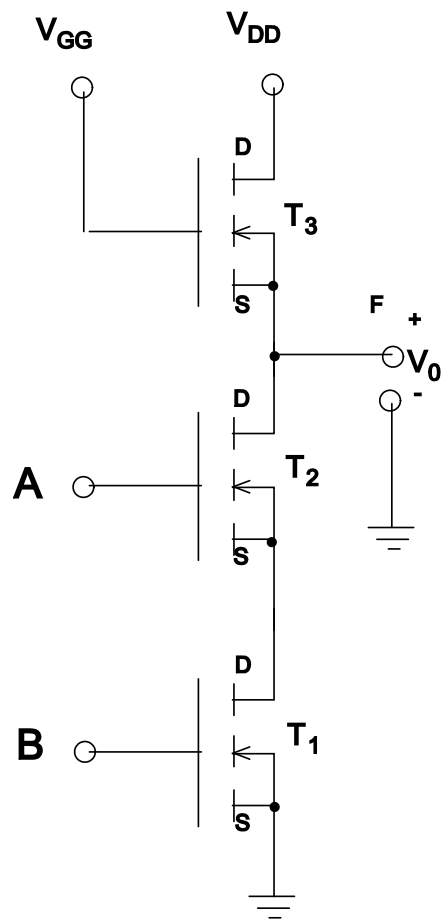
$$V_O = \frac{R_{T1}}{R_{T1} + R_{T2}} V_{DD}$$

For $V_{DD} = 15\text{V}$, $V_o = 0,5\text{V}$

If T_1 off, $V_o = V_{GG} - V_{T2}$, for V_o being approx. V_{DD} , must:

$$V_{GG} = V_{DD} + V_T$$

Static NAND Gate

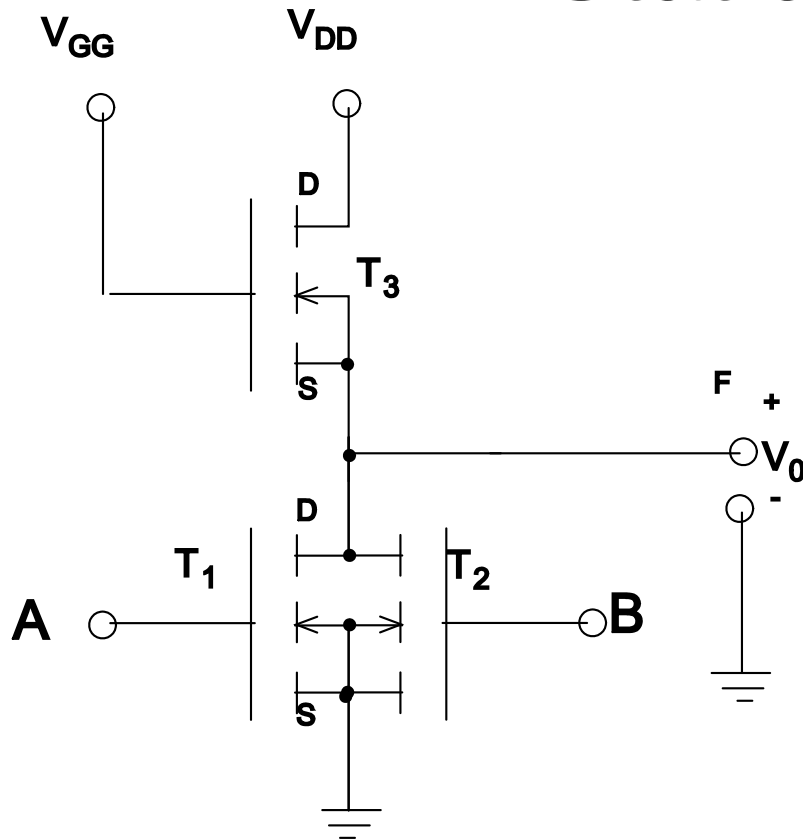


A	B	Q
0	0	1
0	1	1
1	0	1
1	1	0

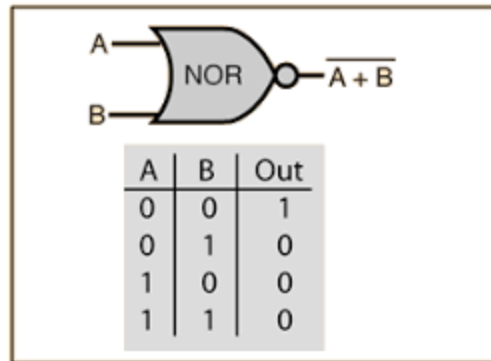
- T_1 and T_2 connected serially, the logic inputs are applied on their gates circuits
- T_3 load resistance
- For making 'better' output voltage levels, mainly the low level very close to 0V, the active resistance must be much greater (20 times) the passing resistance of the input transistors
- Not recommended to serially connect too many input transistors, because the load resistance would become too important and the switching times would grow, damaging the gate's dynamic behavior
- If at both inputs applied $V_{IH} = V_{DD}$, T_1 & T_2 on, $V_o \approx 0V$
- If at least, one input has $V_{IL} = 0V$, corresponding transistor(s) goes off and $V_o \approx V_{DD}$

$$F = \overline{AB}$$

Static NOR Gate

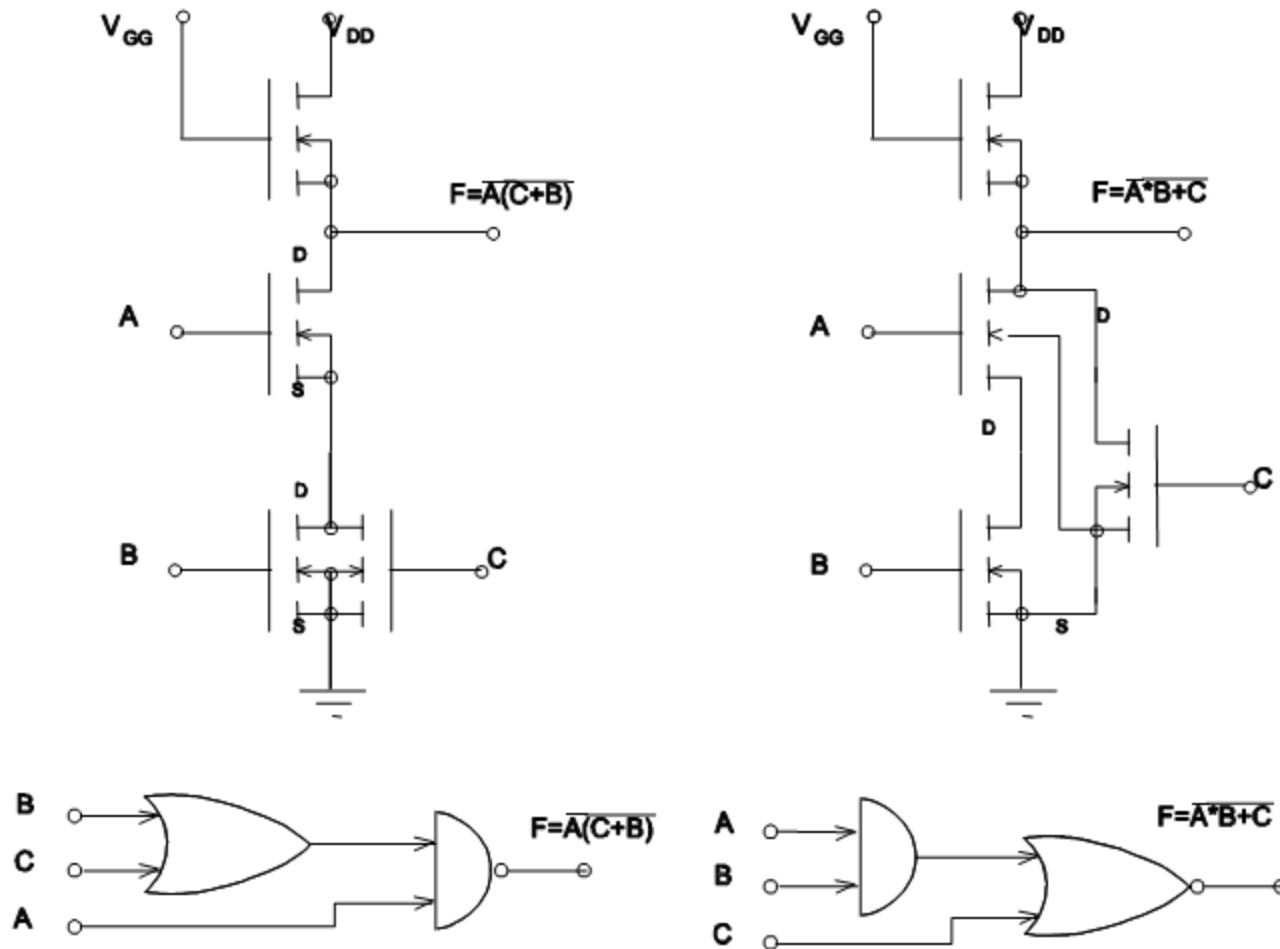


- T_1 & T_2 connected in parallel, their gates are the input circuits
- T_3 acts as load resistance
- Connecting input transistors in parallel doesn't affect the load resistance, so the number of circuit inputs isn't bounded by dynamic considerations
- If both inputs have: $V_{IL} = 0V$, T_1 & T_2 off, $V_o \approx V_{DD}$
- If applying at least at one input $V_{IH} = V_{DD}$, that input transistor is on, $V_o \approx 0V$



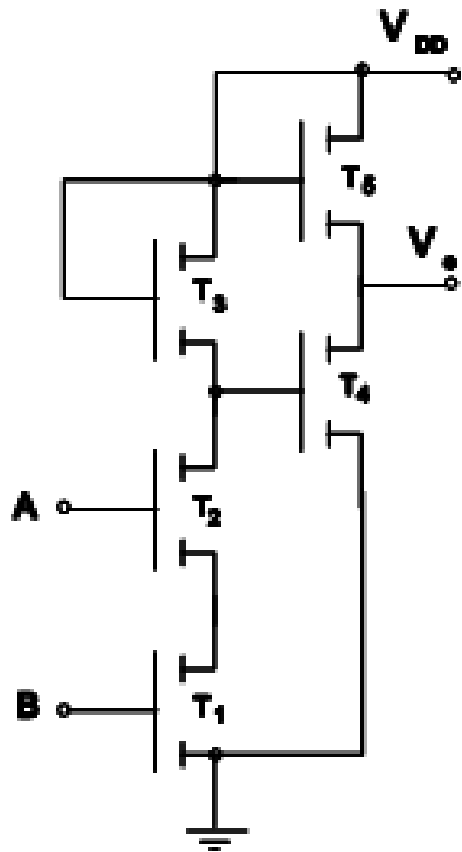
$$F = \overline{A + B}$$

Implementing logic functions

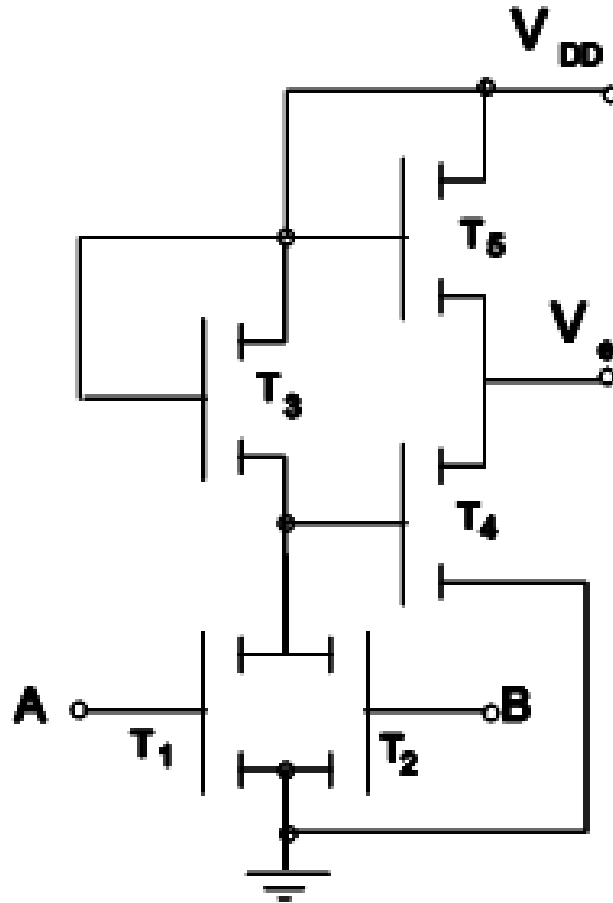


- Using the serial and parallel MOS transistors connections, complex logic functions may be implemented, with a simple structure of the integrated circuit; see behind

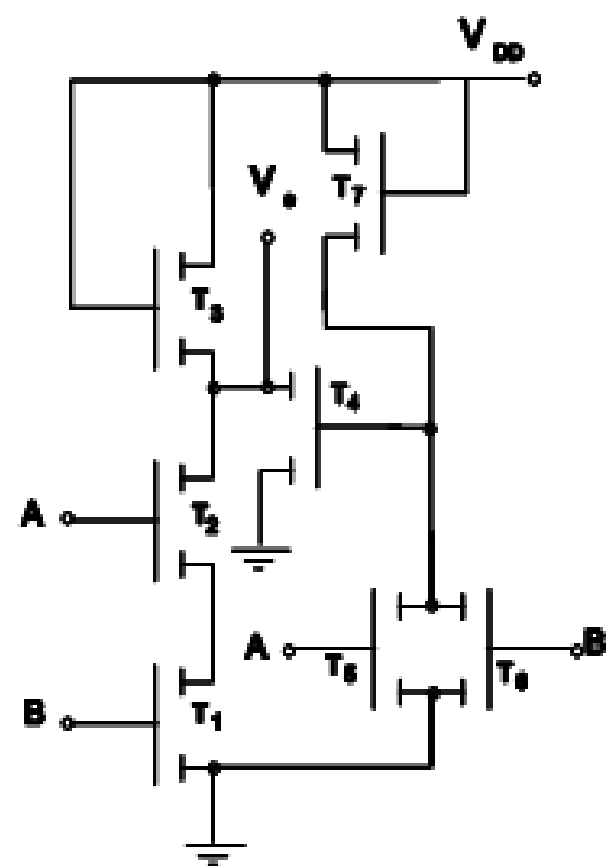
Static AND, OR & XOR Gates



SI



SAU



SAU-Exclusiv

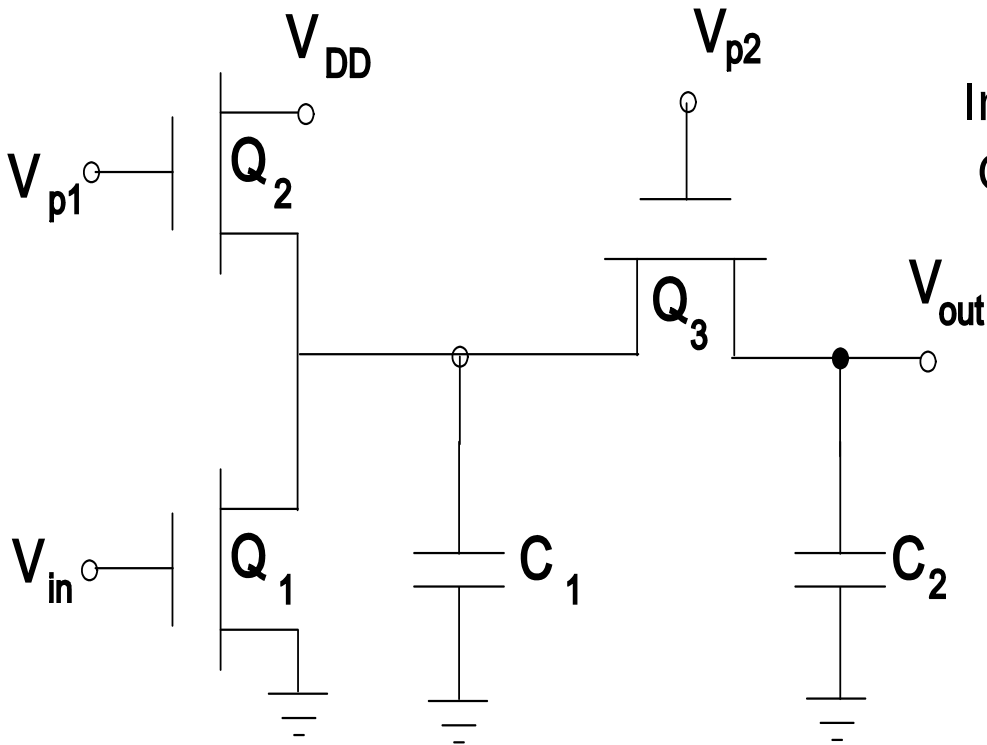
AND & OR gates are built up by inverting the signal from the output of NAND, respectively NOR gates, using an extra inverter, made with transistors T_4 - T_5

For XOR gate: V_0 ="0" for two cases: if T_1 & T_2 are on (A & B inputs both "1") or if T_5 & T_6 off (A & B inputs both "0") (in this case in the gate of T_4 there is a voltage approx. V_{DD} making the transistor on)

Dynamic NMOS circuits

- A basic method to store (memorise) the logic values is to use the input capacitances of the MOS transistors
- A capacitance with no stored charge (discharged) is said to represent a logic '0', respectively a charged capacitance is said to represent a logic "1"
- Signals are applied in the gate circuit, from one capacitor to the other, using transistors driven in conduction by special driving signals (command pulses)
- Operate in a small dissipation power regime
- Dynamic MOS circuits offer a better integration density than the static ones
- Transistors performance doesn't depend on their geometry
- Drawback: more driving (command) signals, more logic

Dynamic NMOS Inverter



Inverter built up from transistors
Q1 & Q2, together with capacitance C1

Output circuit made from transistor Q₃ and storage capacity C₂

There are clock signals:

V_{p1}, applied on Q₂ gate, sampling input value V_{in}, then inverted by Q₁ and stored (memorised) by C₁

V_{p2}, applied on Q₃ gate, making it open and 'copying' the stored charge from C₁ to C₂

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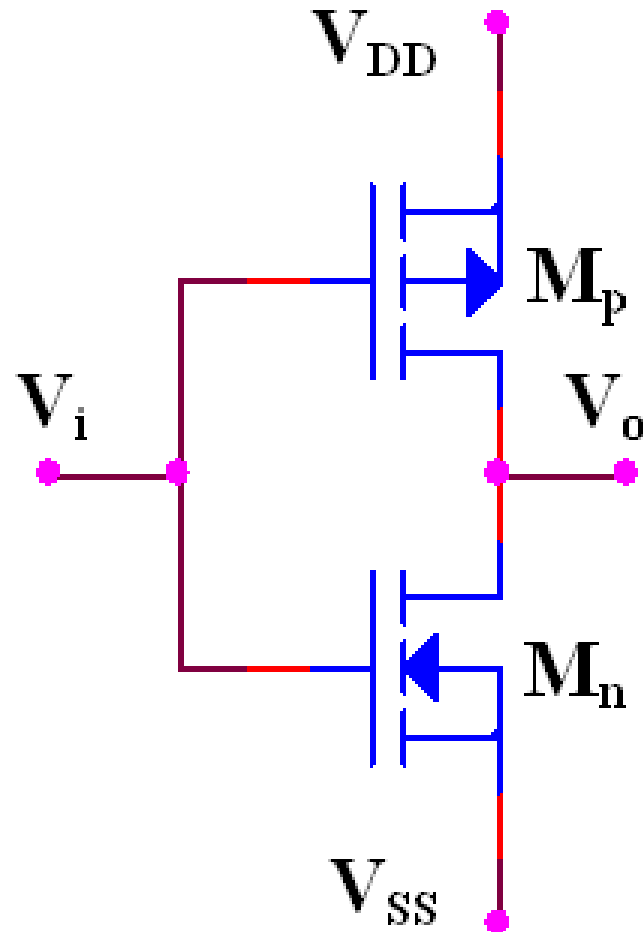
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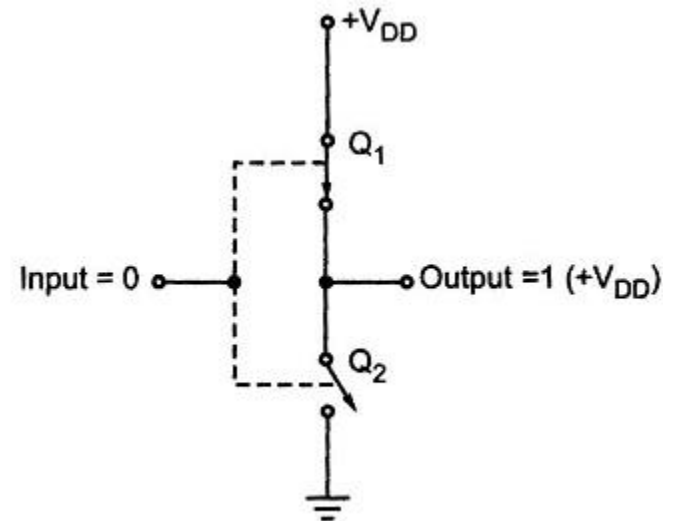
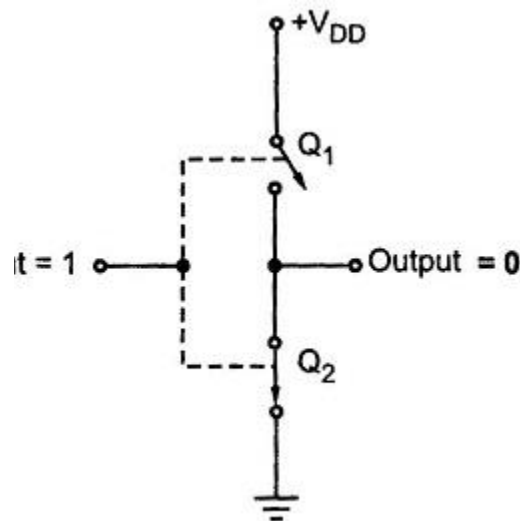
Introduction

- Symetrical and complementary metal-oxide-semiconductor structures
- Present parameter values close to ideal ones
- Reduced **power dissipation** (100 nW in static regime, per gate)
- Broad range for **power supply** voltage: 3-15V sau 3-18V and less (for future)
- Huge **fan-out** (over 100 in static regime)
- in dynamic regime, because each CMOS input presents a 5pF load capacity, there is a trade-off between delay (speed) and number of driven loads
- Broad range of allowed ambiental **temperatures** ($-40^{\circ}\text{C} \div +85^{\circ}\text{C}$)
- **Output voltage** levels very close to 0V for logic '0' state, and respectively close to power supply voltage for logic '1'

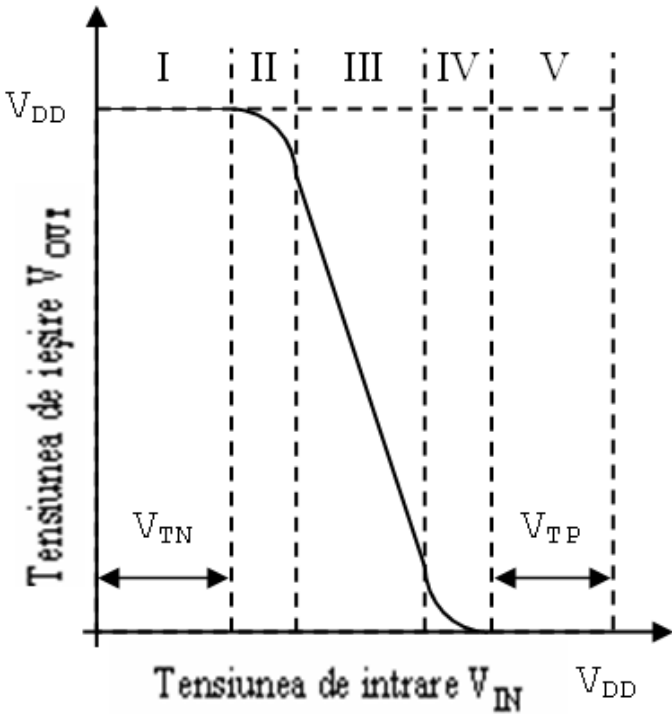
CMOS Inverter



- Pair of MOS transistors, one with **n** channel and one with **p** channel
- $V_i = V_{DD} = '1'$, M_n open, M_p off, $V_o = V_{SS} = '0'$
- $V_i = V_{SS} = '0'$, M_n off and M_p open, $V_o = V_{DD} = '1'$



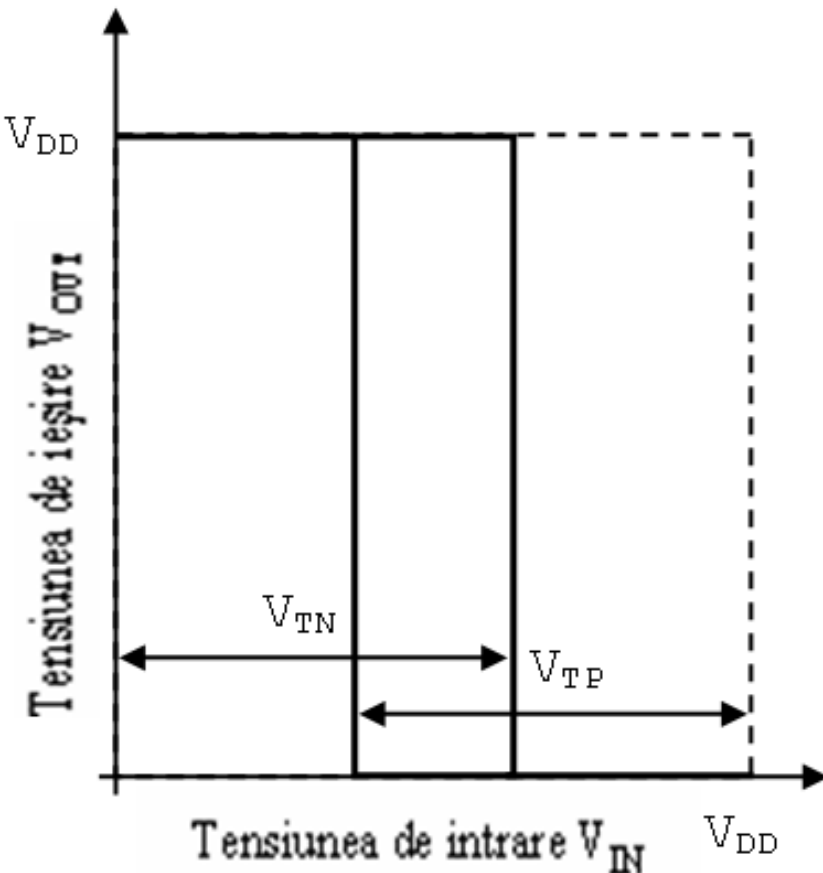
Static Transfer Characteristic



- depends on power supply V_{DD}
- Five operation regions
- V_{TN} threshold voltage for M_n
- V_{TP} threshold voltage for M_p

TENSIUNEA DE INTRARE V_{IN}	REGIUNEA	M_p	M_n
$0 \leq V_{IN} \leq V_{TN}$	I	LINIAR	BLOCAT
$V_{OUT} - V_{TP} \geq V_{IN} \geq V_{TN}$	II	LINIAR	SATURAT
$V_{OUT} - V_{TP} \leq V_{IN} \leq V_{OUT} + V_{TN}$	III	SATURAT	SATURAT
$V_{OUT} + V_{TN} \leq V_{IN} \leq V_{DD} - V_{TP} $	IV	SATURAT	LINIAR
$V_{DD} - V_{TP} \leq V_{IN} \leq V_{DD}$	V	BLOCAT	LINIAR

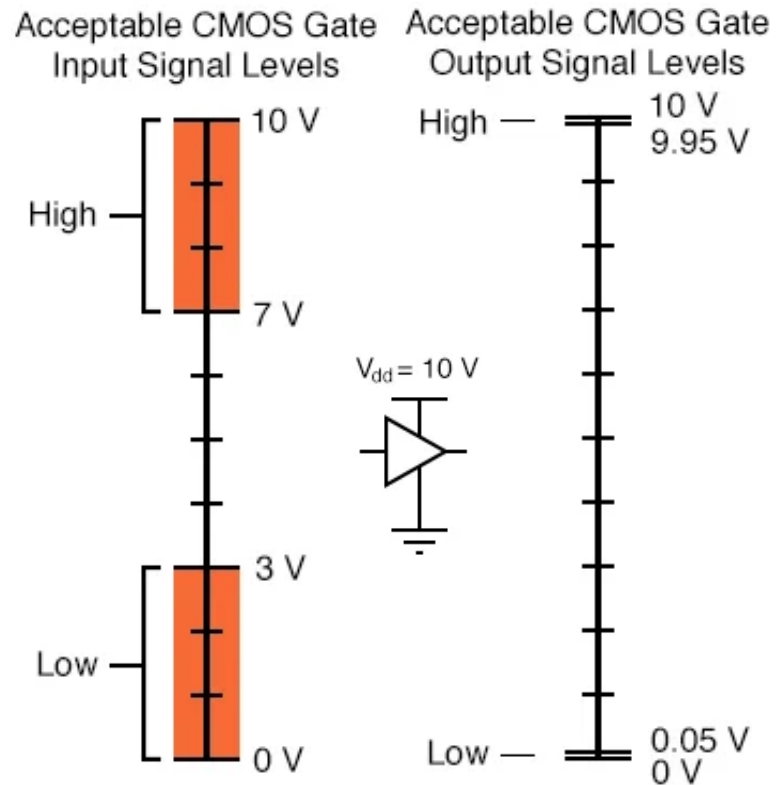
Minimum power supply voltage



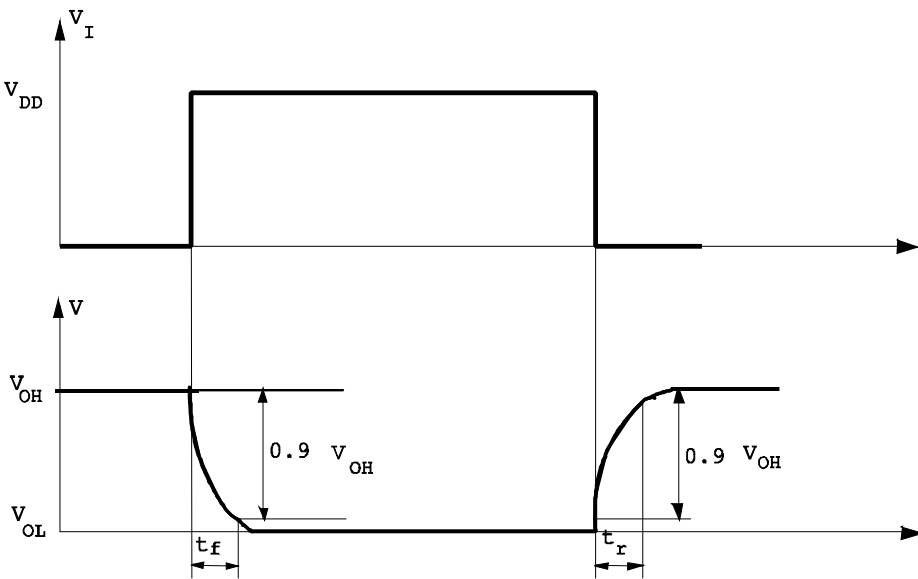
- If V_{DD} lower than:
 $V_{DDmin} = V_{TN} + |V_{TP}|$, inverter will present a hysteresis like transfer characteristic and the gate will NOT behave as a logic gate
- Typical threshold voltage value for standard CMOS: $V_{TN} = |V_{TP}| = 1,5V$
- $V_{DDmin} = 3V$

Voltage Levels and Noise Margins

- $V_{OHmin} = V_{DD} - 0.5V$ (typical: $V_{DD} - 0.01V$)
- $V_{OLmax} = 0.05V$ (typical: $0.01V$)
- $V_{IHmin} = 70\%V_{DD}$
- $V_{ILmax} = 30\%V_{DD}$
- $M_{ZL} = V_{ILmax} - V_{OLmax} = 30\%V_{DD}$
- $M_{ZH} = V_{OHmin} - V_{IHmin} = 30\%V_{DD}$
- Practically, noise immunity is 45...50% from power supply voltage



Circuit Response at an Ideal Pulse



- Circuit has a load capacitance C_S
- t_f and t_r : duration of fall down and raise-up edges for output

- Factors influencing the switching speed:
 - Power supply voltage
 - Circuit configuration (with or without separation circuits for outputs)
 - technology
 - Load capacitances

Power Dissipation

- In **static regime**, one out of two transistors are off; no power consumption, except due to stray currents flowing through megaohms resistors (resistance equivalent to blocked transistors)
- During the **dynamic regime**, each transition will increase consumption, due to:
 - Both complementary MOS transistors are in conduction
 - Parasitic capacitances from circuit's output present charge/discharge cycles

Power Dissipation

- $P_t = P_{cc} + P_{dc} + P_{df}$
- P_{cc} means static power, dissipated by circuit during a steady state, and is due to residual (stray) currents of a locked transistor
- P_{dc} means dynamic power, due to charge/discharge process of parasitic output capacitances
- P_{df} means dynamic power dissipated during transistions, on raising and falling edges, when both transistors are in transitory states
- P_{cc} , is small (nW), because the stray currents are small; they depend on the power supply and temperature (double for any 10 Cgrades)

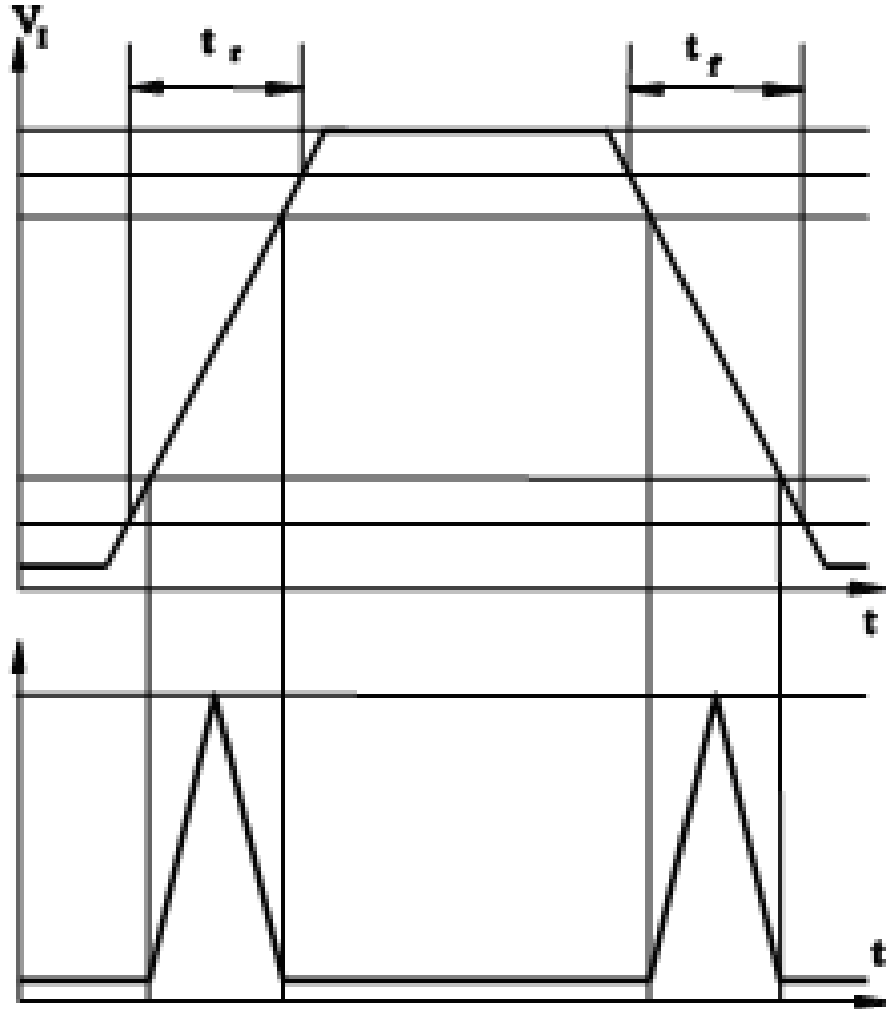
Power Dissipation

$$P_{df} = V_{DD} \cdot \frac{1}{2} \cdot I_{DDmax} \cdot \frac{\text{Durata frontului}}{\text{Perioada semnal}}$$

- I_{DDmax} current due to circuit switching, not considering the charge/discharge currents on parasitic capacitances
- Transistors are on simultaneously during a time period given by moments when signal varies between V_T and $V_{DD}-V_T$, where V_T is the threshold voltage:

$$\frac{\text{Durata frontului}}{\text{Perioada semnal}} = \frac{V_{DD} - 2 \cdot V_T}{V_{DD}} \cdot \frac{t_r + t_f}{T}$$

$$P_{dc} = \frac{C_{PD} \cdot V_{DD}^2}{T} = C_{PD} \cdot V_{DD}^2 \cdot f$$



Power Dissipation

- For a simpler calculus, P_{df} may be considered equal with power absorbed during charge/discharge process of an equivalent capacitance C_{PD} , being summed with the parasitic capacitance from the circuit output, i.e.:

$$P_t = C_{sarcina} \cdot V_{DD}^2 \cdot f + C_{PD} \cdot V_{DD}^2 \cdot f + P_{CC} = (C_{PD} + C_{sarcina}) \cdot V_{DD}^2 \cdot f + I_{rezidual} \cdot V_{DD}$$

- dynamic power-dissipation capacitance, defined as the internal

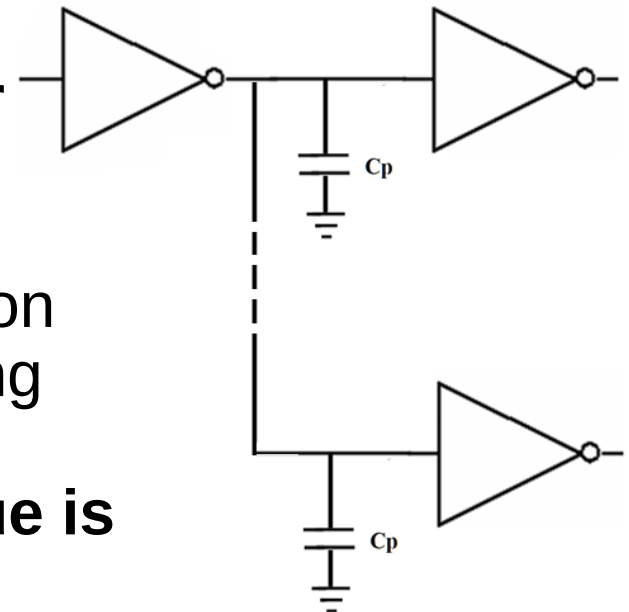
Fan-in, Fan-out

- High input impedance, low input current (10pA)
- One important component of input current is given by the charging/discharging process for CMOS structures input capacitances; during switching the static input capacitance (typical 5pF) is enhanced 5 to 10 times, due to reactions through parasitic capacitances

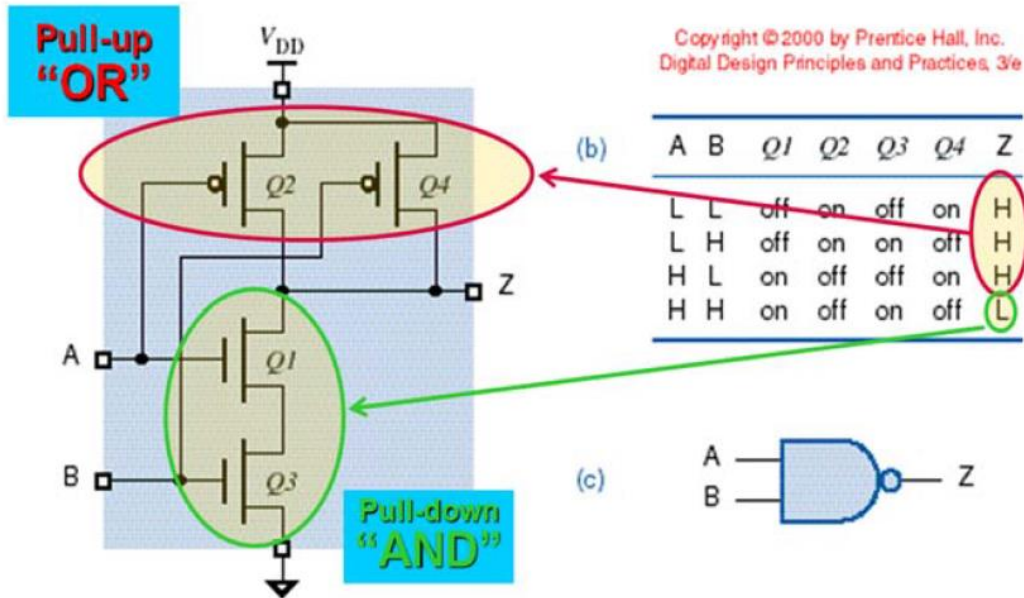
$I_{OL}=0,44\text{mA}$, $I_{OH}= -0,5\text{mA}$ for $V_{DD}=5\text{V}$;

$I_{OL}=0,9\text{ mA}$, $I_{OH}=-0,9\text{ mA}$ for $V_{DD}=10\text{V}$

- Output currents may drive many **similar CMOS** gates.
- Because of capacitive load, depending on number of driven gates, and this affecting performance (propagation delay, power dissipation), in practice the **fan-out value is bounded to 50**

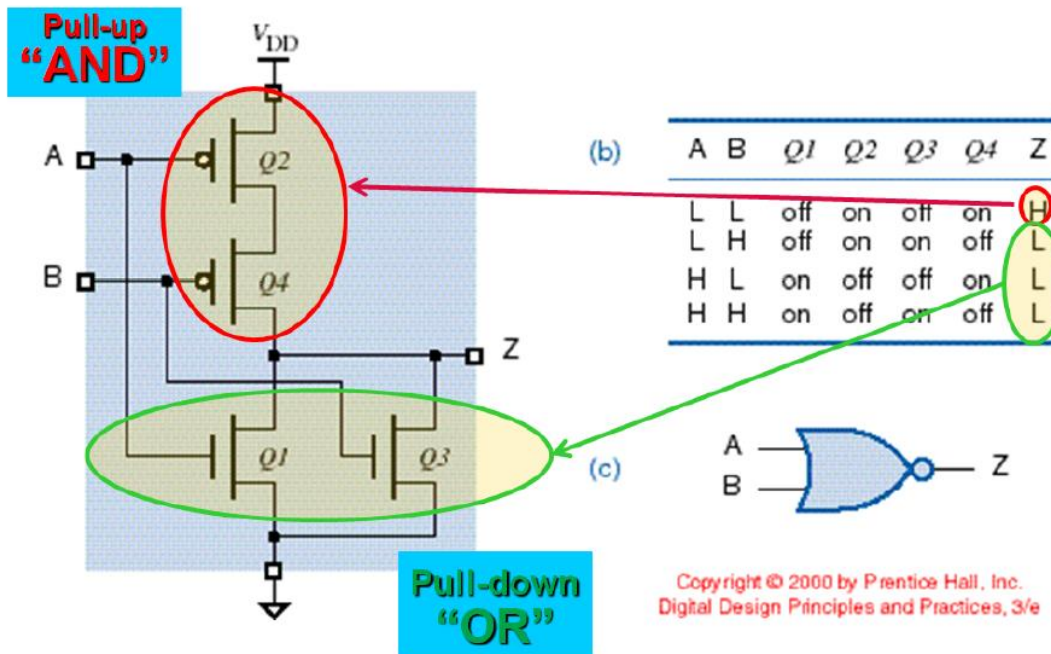


basic CMOS NAND gate



more input combinations
pull output **high** than **low**
→ P-channel pull-ups in
parallel ("OR"),
N-channel pull-downs in
series ("AND")

basic CMOS NOR gate



more input combinations
pull output **low** than **high**
→ P-channel pull-ups in
series ("AND"),
N-channel pull-downs in
parallel ("OR")

HCT Logic Circuits

High-speed CMOS TTL compatible

Used to interface TTLs with CMOS

- Input levels are compatible with TTL logic levels
- CMOS with higher output currents

Power Supply

$$V_{CC}=4,5V\div 5,5V$$

Voltage levels

$$V_{OHmin}=V_{CC}-0,1V$$

$$V_{OLmax}=0,1V$$

$$V_{IHmin}=2V$$

$$V_{ILmax}=0,8V$$

Noise Immunity

$$V_{CC}=4,5V$$

$$M_H=V_{OHmin}-V_{IHmin}=2,4V$$

$$M_L=V_{ILmax}-V_{OLmax}=0,7V$$

Input/output currents

$$I_{OH} = I_{OL} = 4\text{mA}$$

I_{IH} , I_{IL} – negligible ($\approx 10\text{pA}$)

Fan-out/Fan-in

As for CMOS circuits, HCT gate output currents are able to drive a lot of similar circuits

Practically, fan-out is bounded, to keep a high value for the cut-off (maximum) operating frequency

One HCT gate may drive maximum 2 TTL gates

Propagation Times

$$t_{pHL} = t_{pLH} = t_{pd} = 7\text{ns}$$

Static Power Dissipation

$$P_{cc} \approx 10\text{nW}$$

Proposed Problems

- How many 74 series TTL gates may be driven using a HCT gate?
FO=?
 $I_o \text{ (HCT)} = 4\text{mA}$
 $I_i \text{ (TTL)} = 0,04 \text{ mA or } 1,6 \text{ mA}$
 $FO = \min(I_o/I_i) \Rightarrow 4/1,6 \Rightarrow 2$
- What is the maximum value for the resistor mounted between two CMOS gates without modifying the circuit behavior? Is this affecting the noise immunity?

 $I_i \text{ (CMOS)} \text{ very small } (\sim 10\text{pA})$
 $M_{Z(L,H)} = 30\% V_{dd}$

 $\Rightarrow R \text{ can have high values because the low input current}$

Proposed Problems

- Design a circuit using CMOS gate to drive a LED. Following values are considered for this LED: $V_{LED}=1,6V$ and $I_{LED}=20mA$.

If $I='0'$ $\rightarrow$ Q_1 on $\rightarrow$ LED on

If $I='1'$ $\rightarrow$ Q_1 off $\rightarrow$ LED off



A	B	Q
0	0	1
0	1	0
1	0	0
1	1	0

$$R_E = \frac{V_{OH \min} - V_{BE} - V_{LED}}{I_{LED}} \approx 110\Omega$$

